SAN BARTOLOMÉ JHS LA MERCED	TEACHING-LEARNING PLAN	PGF-02-R39 DATE August 12 – November 5, 2026

1. Identification

Area of knowledge: Global Economics
Teacher's name: Nicolas Lopez
Grade: 11
Term: First Term, 2026–2027
Plan code and version: PGF-02-R39, Versión 08
Language: English

2. Learning Objective

LEARNING OBJECTIVE

Analyze uncertainty in Colombian and global economic and operational settings by defining random variables and supports; classifying discrete and continuous cases; interpreting PMFs, PDFs, and CDFs; and recognizing appropriate distribution families. Construct and normalize uniform, triangular, linear, and piecewise densities and interpret continuous probability as area; model periodic phenomena with circular probability, circular means, and resultant concentration; and apply the Poisson-rate/exponential-waiting connection, expectation, and memorylessness while evaluating assumptions. As retained extensions, calculate standard-normal probabilities, standardize contextual variables, use inverse-normal reasoning for cutoffs and intervals, create interactive density visualizations, fit and compare models to data, and communicate justified conclusions through a shareable digital model and an evidence-based APA-style report.

ASSESSMENT CRITERIA

C1. Classify random variables as discrete or continuous; identify support and interpret PMFs, PDFs, and CDFs in economic contexts. *Current Term 1.*
C2. Normalize continuous densities and calculate probabilities geometrically for uniform, triangular, linear, and piecewise models. *Current Term 1.*
C3. Model periodic variables using circular geometry and circular statistics. *Current Term 1.*
C4. Apply and evaluate exponential waiting-time models. *Current Term 1.*
C5. Calculate standard-normal probabilities with a cumulative z-table. *Retained extension; adaptation required.*
C6. Standardize contextual normal variables and calculate probabilities. *Retained extension; adaptation required.*
C7. Use inverse-normal reasoning for percentiles, cutoffs, and central intervals. *Retained extension; adaptation required.*
C8. Build interactive continuous-density models with shaded probability regions. *Retained project extension; adaptation required.*
C9. Fit and compare probability models to data. *Retained project extension; adaptation and data resolution required.*
C10. Communicate a sourced comparative analysis with methods, evidence, conclusions, and APA references. *Retained project extension; adaptation required.*

3. Conceptual standards of the disciplinary knowledge that supports the narrative's development

1. A random variable maps outcomes to numbers. Its support lists allowable values. A PMF assigns mass to countable values; a PDF is a nonnegative density whose interval area is probability; a CDF, $F(x) = P(X \leq x)$, accumulates probability. Classify by the variable and support, not merely by context.
2. A valid PDF is nonnegative and has total area one. Normalize constants before finding probabilities; use rectangles, triangles, trapezoids, complements, and sums across all piecewise breakpoints. Density height is not itself probability and may exceed one.
3. Periodic supports identify cycle endpoints. Represent wrapped intervals as disjoint arcs; use sectors or circular segments when the supplied density graph requires them. For observations θ_i, use mean unit-vector components $\bar{C}$, $\bar{S}$, circular mean $\operatorname{atan2}(\bar{S}, \bar{C})$, and resultant length $R = \sqrt{\bar{C}^2 + \bar{S}^2}$ to report direction and concentration.
4. Under independent events at a stable Poisson rate λ, the next waiting time has $T \sim \text{Exp}(\lambda)$, $F(t) = 1 - e^{-\lambda t}$, $P(T > t) = e^{-\lambda t}$, and $E(T) = 1/\lambda$. Memorylessness, $P(T > s + t \mid T > s) = P(T > t)$, is conditional on an unchanged mechanism and rate.
5. For $Z \sim N(0, 1)$, the cumulative table gives $\Phi(z) = P(Z < z)$. Symmetry $\Phi(-z) = 1 - \Phi(z)$, complements, subtraction for intervals, and addition of disjoint tails determine left, right, central, and two-tail regions.
6. A contextual normal model $X \sim N(\mu, \sigma^2)$ is converted with $z = (x - \mu)/\sigma$. Retain original units in the interpretation and check that the normal assumptions are plausible.
7. Inverse-normal reasoning begins with a cumulative probability p, obtains z_p satisfying $\Phi(z_p) = p$, then converts with $x = \mu + z_p\sigma$. Translate upper-tail and central-percent statements into cumulative probabilities before finding cutoffs.
8. An interactive PDF must expose valid parameter controls, preserve non-negativity and normalization, label support and density, and shade the correct below, above, or between region while reporting its area.
9. Data-based modeling requires identifying and cleaning a continuous variable, estimating each candidate's parameters, comparing a histogram with fitted curves, checking support and shape, and interpreting fit and probabilities without claiming that visual agreement proves a model.
10. Evidence-based reporting documents purpose, data and sources, methods, estimates, selected probabilities, screenshots, model comparisons, conclusions, limitations, and reproducible AI-assisted work, with in-text citations and APA references.

4. Pre-lesson and Context: C1 Discovery Opening

Framework	A 55-minute inquiry cycle: *notice and name* a variable, *sort and justify* its type/model, *represent* support and probability, then *revise* after whole-class critique. This activates prior knowledge before explicit instruction rather than treating the solved sheet as a test.
Grouping and resources	Pairs rotate through four stations; each pair uses context cards adapted from the full current C1 activity, a support-number-line sheet, PMF/PDF/CDF cards, colored pencils, and one teacher-controlled copy of the compact solved handout for feedback. OPEN/Teams may distribute accessible copies.
Timing and products	Launch 5 minutes; four 8-minute stations; synthesis 13 minutes; transition 5 minutes. Each pair submits four variable/support cards, a family-sort record, one corrected misconception, and an individual exit response.
Vocabulary	outcome, random variable, support, discrete, continuous, probability mass, density, cumulative probability, PMF, PDF, CDF, Bernoulli, binomial, geometric, Poisson, uniform, triangular, linear, normal.
Command words	identify, define, classify, distinguish, represent, match, justify, interpret, compare, correct.

Station 1 – Name the variable and its support

Students distinguish a numerical random variable from the underlying event. They write supports for a single transfer clearing (Bernoulli, $\{0, 1\}$ after coding), passes among 40 inspections (binomial, $\{0, \ldots, 40\}$), calls through the first appointment (geometric, $\{1, 2, \ldots\}$ under the activity's trial-count convention), and continuously measured delivery or price intervals. **DUA support:** sentence frame "Let X be $\ldots$; possible values are $\ldots$" and pre-drawn discrete dots/continuous number lines. **Extension:** explain why a recorded rounded measurement can be discrete while the modeled quantity remains continuous.

Station 2 – Mass, density, and accumulation

Pairs match a PMF to countable point masses, a PDF to nonnegative area, and a CDF to $P(X \leq x)$. They reject "$f(x)$ is the probability of exact continuous value x" and state that $P(X = x) = 0$ for a continuous variable although $f(x)$ may be positive. **DUA support:** color-code mass, area, and accumulated area; provide labeled axes. **Extension:** sketch how jumps in a discrete CDF differ from a continuous CDF.

Station 3 – Recognize discrete families

Students use structural clues, verified in the current worked material: one binary trial is Bernoulli; successes in a fixed number of independent equal-probability trials are binomial; trials through first success are geometric; separate events in a fixed interval at approximately stable rate are Poisson. Current Colombian contexts include Bogotá transfers, venture calls, inspections, enquiries, sustainable-trade errors, supplier checks, finance approvals, tourism reports, and buyers. **DUA support:** a four-question checklist (one trial? fixed n? stop at first success? count in interval?). **Extension:** identify which independence or constant-rate assumption could fail.

Station 4 – Recognize continuous families

Students classify verified activity cases: bounded equal plausibility as uniform; stated bounds and interior most likely value with linear rise/fall as triangular; one straight increasing/decreasing density as linear; approximate symmetry and taper around a center as normal. Contexts include event budgets, export measurements, platform times, food weights, finance/content timing, tourism/logistics durations, design measurements, technology prices, commerce, and sustainability moisture. **DUA support:** shape cards and a bounded/symmetric/mode decision tree. **Extension:** explain why "bounded and increasing" is not normal and why a triangular density needs two pieces.

Whole-class synthesis	Build a public concept map: outcome $\rightarrow$ variable $\rightarrow$ support $\rightarrow$ discrete/continuous $\rightarrow$ PMF/PDF $\rightarrow$ CDF. Reveal only the corresponding compact-handout classifications after justifications, let pairs correct in another color, and emphasize that the handout is reinforcement, not a separate assessment.
Transition to N1	Ask: "Once a continuous family and support have been recognized, how can its unknown height become a valid probability model?" This question launches C2 while C1 vocabulary remains in use.

5. Experience: Three Pedagogical Actions

Pedagogical Action N1 – From Random Variables to Continuous and Waiting-Time Models	
Criteria and status	C1–C4; **current Term 1 core sequence** and therefore the principal instructional action.
Supported cycles/dates/hours	Instructional window supported by current materials: August 12–November 5, 2026. The repository does not assign cycle numbers, criterion dates, or hours; these remain to be scheduled within the institutional calendar.
Deadline	
Role in sequence	Establish model language (C1), construct area-based densities (C2), transfer area reasoning to periodic settings (C3), and connect event rates to waiting-time models and assumption checks (C4).
Exact materials	Current C1, C2, C3, and C4 practice activities; current cumulative C1–C4 assessment-style solved set; compact solved C1 handout as reinforcement only.
Vocabulary	random variable, outcome, support, PMF, PDF, CDF, normalization, breakpoint, complement, periodic, wrapped interval, sector, circular segment, circular mean, resultant, Poisson rate, exponential, survival, expectation, memorylessness, suitability.
Command words	define, identify, classify, construct, normalize, calculate, decompose, interpret, compare, justify, assess.

C1 – Variables, representations, and distribution recognition	
Objective and format	Through teacher think-aloud, card sort, paired classification, and individual explanation, define the variable/support; decide discrete or continuous; distinguish PMF, PDF, and CDF; and recognize Bernoulli, binomial, geometric, Poisson, uniform, triangular, linear, and normal structure in the activity's Colombian economic contexts.
Theory and notation	For discrete X, $p_X(x) = P(X = x) \geq 0$ and $\sum_x p_X(x) = 1$. For continuous X, $f_X(x) \geq 0$, total area is one, and $P(a \leq X \leq b)$ is area; $F_X(x) = P(X \leq x)$ in either case. Support must be stated explicitly.
Procedure	Underline the measured quantity; name X and units; list or interval-mark support; test countability; choose PMF/PDF; identify structural clues; name a family only when its defining conditions fit; justify in one context sentence.
Representative source problems and verified logic	Use the full activity's four discrete groups and four continuous groups. Verified examples include one transfer (Bernoulli), passes among 40 inspections (binomial), calls until first appointment (geometric), events in a fixed hour (Poisson), bounded equally plausible platform time (uniform), bounded interior mode (triangular), a single straight changing density (linear), and symmetric tapering measurement (normal). The current cumulative set confirms loan default, sales calls, supplier delivery time, workshop time, invoice errors, arrivals, processing time, and spending classifications.
Common errors	Classifying the context rather than the variable; calling every count Poisson; confusing fixed-n binomial with stop-at-first-success geometric; treating PDF height as point probability; omitting support; calling every bell-like or bounded graph normal.
Practice and assessment	Students annotate and justify selected cases from each current C1 group, then answer cumulative classification prompts without the key. The compact solved handout is used afterward for correction and verbal retrieval, never as a separate assessment.
DUA support and extension	Support: recognition checklist, word bank, labeled supports, mass-versus-area diagram, and worked/partially worked example. Extension: propose a competing model, identify a violated assumption, or distinguish measured from rounded support.
Student evidence	Variable definition, support, type, representation, family, and written justification for each selected case, plus color-coded corrections.

C2 – Construct and use continuous densities geometrically	
Objective and format	Teacher models one full construction; groups use area-decomposition cards; pairs solve a new context; individuals complete a transfer problem and interpretation.
Theory and notation	A PDF is nonnegative and normalized: $\int f_X(x)\,dx = 1$. On straight pieces, integral area is a rectangle, triangle, or trapezoid. Adjacent formulas must honor stated continuity; a jump is permissible in piecewise-uniform models. Include $f_X(x) = 0$ outside support and split events at every breakpoint.
Required eight-step routine	1. Define the variable. 2. Identify support. 3. Infer density shape. 4. Establish non-negativity. 5. Normalize total area to one. 6. Write the complete PDF. 7. Identify the requested interval. 8. Calculate and interpret its area.
Exact source activities and verified problems	Use all five current C2 sections: microcredit disbursement $T \in [2, 10]$ gives $h = 1/8$, $P(T > 7) = 3/8$, and $P(3 \le T \le 7) = 1/2$; coffee-auction premium gives $f_X(x) = (x + 2)/48$ on $[0, 8]$, with tail $3/8$ and middle probability $1/2$; last-mile order volume has support $[2, 10]$, mode 5, peak $1/4$, $P(Q > 7) = 9/40$, and $P(4 \le Q \le 8) = 11/15$. Merchant reconciliation uses breakpoints 3 and 8, heights $1/15, 2/15, 1/30$, tail $2/15$, and interval $23/30$. Weekly payout liquidity uses breakpoints $2, 5, 7$, maximum $4/25$, tail $11/50$, and central probability $68/75$.
Verified solution logic	Translate verbal ratios into heights first; sum whole-support rectangles/triangles/trapezoids to solve the constant; derive each line from two points; test meeting heights at continuous breakpoints; shade the event; then add pieces or subtract simpler tails. Endpoints do not change continuous probability.
Common errors	Normalizing only the requested region; forcing a triangular formula onto a positive-endpoint line; omitting zero outside support; failing to split at a breakpoint; confusing height with area; using a complement without accounting for both tails.
DUA support and extension	Support: pre-drawn/shaded graphs, partially labeled breakpoint heights, formula/area sheet, and an eight-box template. Extension: derive segment equations, compare direct area and complement, change a height ratio while preserving normalization, or diagnose discontinuity/non-negativity.
Student evidence	Complete piecewise PDF, normalization equation, labeled/shaded graph, area calculation, probability, and contextual interpretation for each model family.

C3 – Periodic probability and circular statistics	
Objective and format	Use clock/compass manipulatives, guided circular geometry, calculator-supported vectors, and an individual circular-versus-linear model defense.
Theory and notation	On one cycle $f(\theta) = f(\theta + 2\pi)$ and total probability is one. A wrapped event is split across 0. Circular uniform probability is arc width divided by 2π (or 360°). Compute $\bar{C} = n^{-1} \sum \cos\theta_i$, $\bar{S} = n^{-1} \sum \sin\theta_i$, $\bar{\theta} = \operatorname{atan2}(\bar{S}, \bar{C})$, and $R = \sqrt{\bar{C}^2 + \bar{S}^2}$. The cumulative solved evidence additionally normalizes semicircular-arc densities and uses semicircle/sector-minus-triangle circular-segment areas; a circular-looking PDF graph alone does not make its variable periodic.
Procedure	Identify the repeated cycle and units; choose a one-cycle support; map or split the event; normalize the supplied density/arc; calculate arc, sector, or segment area; for data, average unit vectors and interpret both angle and R; justify circular versus linear treatment.

Exact source activities and verified results	Current C3 problems verify $P(\pi/3 \leq \Theta \leq 5\pi/6) = 1/4$; within 30° of north is $[330^\circ, 360^\circ) \cup [0^\circ, 30^\circ]$ with probability $1/6$; directions $10^\circ, 350^\circ$ yield circular mean 0° and $R \approx 0.9848$ rather than arithmetic mean 180°; $0^\circ, 120^\circ, 240^\circ$ yield $R = 0$ and no unique mean; and 22:30–01:30 has probability $3/24 = 1/8$ under a circular-uniform time model. Use the current cumulative set's semicircular density problems for sector/circular-segment transfer.
Common errors	Treating 330° to 30° as empty; averaging angles arithmetically across zero; reporting $\operatorname{atan2}(0, 0)$ as an angle; interpreting low R as computational failure; confusing a circular support with an arc-shaped density; omitting the triangle subtraction in segment area.
DUA support and extension	Support: clock/compass overlays, radian-degree conversion strip, pre-split wrapped arcs, vector-component template, and labeled sector diagrams. Extension: solve intervals spanning zero, derive an arc-density scale factor, compare circular uniform with von Mises/wrapped-normal candidates, or discuss whether peak-demand time supports a single mean.
Student evidence	One wrapped-event representation and probability, one normalized arc/segment calculation, circular mean and R with interpretation, and a model-choice paragraph.

C4 – Exponential waiting-time models and suitability

Objective and format	Teacher rate-conversion demonstration, paired probability cases, conditional-probability comparison, and group assumption audit followed by an individual operational recommendation.
Theory and notation	$T \sim \operatorname{Exp}(\lambda)$ on $t \geq 0$ has $f(t) = \lambda e^{-\lambda t}$, $F(t) = 1 - e^{-\lambda t}$, survival $e^{-\lambda t}$, $P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}$, $E(T) = 1/\lambda$, and $P(T > s + t \mid T > s) = P(T > t)$. A stable independent Poisson event process motivates, but does not automatically validate, exponential interarrival waits.
Procedure	Define the next-event wait and unit; estimate or infer λ; convert rate to the wait's unit; select CDF, survival, difference, expectation, or memoryless expression; substitute and interpret; inspect rate stability, independence, scheduling, queues, and dependence/aging before recommending the model.
Exact source activities and verified logic	Use fintech verification: 144 enquiries in 12 hours gives 12/hour $= 0.2$/minute and mean wait 5 minutes. Use off-peak Chapinero courier requests for below 10, above 20, and between 10–20 minute probabilities; tourism enquiries to show conditional calculation and memorylessness; scheduled buses, payment confirmations, and aging scanners to audit suitability; flower-export logistics to infer rate from mean and qualify Valentine's-week use; and digital-wallet alerts to infer λ from a reported survival probability and calculate conditional next-hour/interval probabilities.
Common errors	Mixing hourly λ with minutes; using $e^{-\lambda t}$ for "within" rather than "exceeds"; subtracting CDFs in reverse; treating $1/\lambda$ as a guaranteed wait; adding elapsed time under memorylessness; assuming schedules, changing rates, queues, coordinated fraud, or aging are exponential.
DUA support and extension	Support: rate-unit ladder, formula-choice flowchart, exponential graph with CDF/survival regions, and conditional timeline. Extension: solve λ from a probability, compare piecewise time-block rates, verify memorylessness algebraically, or propose data checks and an alternative model.
Student evidence	Defined variable/support, rate conversion, model, below/above/between and expected-wait calculations, conditional explanation, and evidence-based suitability judgment.
N1 practice and assessment	Practice moves from worked examples to paired source problems and then closed-key retrieval. The current cumulative solved set is held back as an assessment-style rehearsal/feedback source: students first answer a teacher-adapted unsolved copy or selected prompts, then compare methods with its embedded solutions.
Table-free "I can" exit ticket	Students complete four bullets: *I can* classify a variable and justify its family; normalize a PDF and obtain area; represent a wrapped event/circular mean; and select and qualify an exponential calculation. Each bullet includes one symbol or diagram and one context sentence.

Independent study	Rework one corrected problem per criterion without viewing the solution, annotate why the original error occurred, and submit the new reasoning with a self-check against the relevant worked source.
Required evidence	Station record; selected current-activity solutions; graphs and calculations; N1 exit ticket; cumulative response; feedback annotations; and corrected resubmission.
Material readiness and risks	Current activities contain embedded solutions, so student-facing extracts must hide keys. The cumulative file is an assessment-style solved set, not a separate current unsolved exam, and its internal title says Term 3; relabel/adapt before distribution. Shared preamble/graph dependencies lie outside this folder. No criterion calendar or deadline is supplied.

Pedagogical Action N2 – Normal Probability, Standardization and Percentile Cutoffs	
Criteria and status	C5–C7; **retained legacy Term 3 extension, not established as current Term 1**. All student-facing labels and assessment content require Grade 11 First Term 2026–2027 adaptation and review.
Supported cycles/dates/hours	Possible only within August 12–November 5, 2026 if adopted. Repository evidence supplies no cycle, date allocation, or hours.
Deadline	
Role in sequence	Extend normalized continuous area from N1 to a fixed reference model, then translate contextual thresholds to z-scores and reverse cumulative probabilities to contextual cutoffs.
Exact materials	Retained C5, C6, and C7 practice activities; supplementary 4-extra_material/normal-table.tex; legacy C1–C7 midterm question/solution pair used only as a question/solution reference after adaptation.
Vocabulary	standard normal, z-score, cumulative probability, symmetry, complement, left/right tail, percentile, cutoff, central interval, inverse normal.
Command words	shade, read, calculate, standardize, convert, invert, interpret, verify.

C5 – Standard-normal probability	
Objective, format, and theory	In a z-table workshop, shade first and calculate second for $Z \sim N(0,1)$. Read $\Phi(z) = P(Z < z)$; use $\Phi(-z) = 1 - \Phi(z)$, $P(Z > z) = 1 - \Phi(z)$, $P(a < Z < b) = \Phi(b) - \Phi(a)$, and add disjoint tails.
Procedure and representative verified results	Name region; sketch/shade; rewrite in cumulative-left form; read row and hundredths column from normal-table.tex; apply symmetry/complement; calculate and reason-check. The retained practice verifies $\Phi(0.85) = 0.8023$, $P(Z > 1.20) = 0.1151$, $P(0.35 < Z < 1.40) = 0.2824$, and $P(Z < -1.05$ or $Z > 1.20) = 0.2620$.
Common errors, sources, support, extension, evidence	Errors: reading a right tail directly, losing a negative sign, subtracting in reverse, or adding overlapping regions. Source: retained C5 activity and normal table; legacy midterm only after adaptation. Support: labeled curve, table-reading example, identity sheet. Extension: derive identities by symmetry and compare central/two-tail regions. Evidence: shaded diagram, transformed expression, lookup, result, and check.

C6 – Contextual standardization	
Objective, format, and theory	Use guided templates then individual contexts to transform $X \sim N(\mu, \sigma^2)$ with $z = (x - \mu)/\sigma$, calculate the appropriate standard-normal area, and interpret in original units.
Procedure and representative verified results	State X, units, μ, σ; sketch threshold(s); standardize each; apply C5; interpret. Retained worked examples include $P(X > 52)$ for $X \sim N(56, 5^2)$: $z = -0.80$ and probability 0.7881; a central case gives $\Phi(1.15) - \Phi(0.25) = 0.2762$; and a mixed central interval gives 0.6820.
Common errors, sources, support, extension, evidence	Errors: treating variance as σ, reversing $x - \mu$, retaining contextual values in the z-table, or omitting units/meaning. Source: retained C6 activity and adapted midterm reference. Support: standardization template and two-axis diagram. Extension: compare contextual distributions that share a z-score or critique normal plausibility. Evidence: annotated contextual sketch, both z-values, probability, and original-unit interpretation.

C7 – Inverse-normal cutoffs and intervals	
Objective, format, and theory	Translate percentile/tail language into cumulative probability, obtain z_p with $\Phi(z_p) = p$, and return to context with $x = \mu + z_p\sigma$. For a central proportion c, use equal tail probabilities $(1-c)/2$ unless context says otherwise.
Procedure and representative verified results	Sketch; convert upper-tail to $p = 1 - q$ or split central tails; find inverse-normal value; transform; label units and verify the retained area. Retained C7 solutions include top 5% exam cutoff $z_{.95} \approx 1.645$, $x \approx 86.8$; central 95% battery life $[15.1, 24.9]$ hours; and middle 90% tolerance $[9.868, 10.132]$ mm.
Common errors, sources, support, extension, evidence	Errors: using the tail size as a left cumulative probability, confusing a percentile with a probability result, using $\mu + z$ without σ, or assigning all excluded area to one side of a central interval. Source: retained C7 activity and adapted midterm reference. Support: probability-to-z-to-x template and percentile diagram. Extension: derive asymmetric cutoffs or compare how μ, σ affect intervals. Evidence: shaded target area, cumulative conversion, inverse value, cutoff(s), units, and interpretation.
N2 practice and assessment	Prior practice comprises retained worked C5–C7 activities used as examples and newly adapted unsolved items. The second learning evidence is only *planned*: no current unsolved C5–C7 assessment exists. The legacy, mislabeled midterm may supply questions and its paired solutions may guide feedback, but both require label, scope, notation, and suitability review before use.
Table-free "I can" exit ticket	*I can* shade and calculate a standard-normal region; convert a contextual threshold to z and back to meaning; and convert a percentile or central percentage into cutoff(s). Students attach one corrected misconception.
Independent study and required evidence	Complete one left/right/interval/two-tail z-table set, one contextual standardization, and one cutoff/central interval; submit sketches, table entries, calculations, interpretations, feedback, and correction.
DUA and extension across N2	Use formula sheets, pre-shaded curves, enlarged table coordinates, worked negative-z examples, standardization/inverse templates, and calculator checks. Extend through identity derivation, alternate parameters, assumption critique, and forward-probability verification of inverse answers.
Material readiness and risks	All C5–C7 activities and the midterm pair are retained legacy resources. The midterm is internally Grade 10, Term 3, 2025–2026 and cannot be issued unchanged. The normal table only lists positive z values through 2.99, so symmetry and any out-of-range policy must be taught; no current assessment, schedule, or deadline is present.

Pedagogical Action N3 – Interactive Probability Models, Data Fitting and Comparative Reporting	
Criteria and status	C8–C10; **retained legacy project extension, not established as current Term 1**. Adapt all Grade/Term labels and verify technology, data, links, privacy, and deployment before adoption.
Supported cycles/dates/hours	Possible only within August 12–November 5, 2026 if adopted. No cycles, project dates, or hours are established in current materials.
Deadline	
Role in sequence	Make N1/N2 models dynamic, test candidate models against observations, and communicate a reproducible comparison rather than presenting a graph without evidence.
Exact materials	Retained 2-learning_evidences/old_G11_T3_L4_C8C9C10_activity.tex only; its external tutorials, SharePoint links, relative CSV links, hosting service, and AI-record requirements must be checked before release.
Vocabulary	interactive PDF, parameter, slider/control, shaded region, upload, clean, estimate, histogram, fitted curve, candidate model, deploy, limitation, reproducibility, APA citation.
Command words	build, constrain, upload, clean, estimate, fit, compare, validate, deploy, document, cite, conclude.

C8 – Interactive continuous-density visualizations	
Objective, format, theory, and procedure	In a scaffolded coding studio, build uniform, triangular, linear, piecewise, and normal model modes. For each: define valid controls and support; compute/validate parameters and normalization; plot the PDF; allow below, above, and between events; clamp or flag invalid inputs; shade exactly the requested area; display probability and interpretation; test endpoints and parameter changes.
Common errors and representative source task	The legacy brief explicitly requires interactive PDFs and below/above/between shading. Avoid negative densities, unnormalized curves, reversed bounds, shading outside support, and a graph whose reported probability is unrelated to the shaded area.
Support, extension, and student evidence	Support: starter interface, model-specific checklist, validated sample parameters, test protocol, and screenshot guide. Extension: compare numerical and geometric/integral results or create accessible tooltips. Evidence: source files, test log, working shareable output, screenshots for every family/event type, and visible AI-development record if AI was used.

C9 – Fit and compare models to data	
Objective, format, theory, and procedure	In pairs with individual audit notes: upload a verified dataset; identify variable, unit, missing/non-numeric/out-of-range values and cleaning decisions; estimate parameters appropriate to each candidate; plot a density-scaled histogram and fitted curves; compare support, center, spread, shape, and an identified quantitative/graphical criterion; calculate selected probabilities; interpret; test; deploy; document AI prompts, generated fragments, corrections, and verification.
Representative source task and common errors	The retained brief requests the three provided CSVs, histograms, at least three fitted continuous models per dataset, parameter estimates, probabilities, and strengths/limitations. Avoid fitting before cleaning, overlaying a PDF on count-scale bars, forcing impossible support, treating the best-looking curve as proven true, data leakage, undocumented AI code, or claiming inaccessible data were analyzed.
Support, extension, and student evidence	Support: upload validator, cleaning log, parameter-estimation sheet, histogram-scale example, candidate comparison scaffold, and test checklist. Extension: compare multiple candidates with a justified metric, sensitivity to bin width/outliers, or train/validation reasoning. Evidence: data provenance/cleaning record, estimates, histograms, curves, comparisons, probability outputs, code, tests, deployed/shareable link, and AI-work record.

C10 – Evidence-based comparative report	
Objective, format, theory, and procedure	Individually or with clearly attributed group contributions, produce an APA-style report: modeling purpose/question; data and sources; variable and cleaning; methods and formulas; candidate models and parameter estimates; screenshots of interactive/fitted results; selected probability questions and results; comparative evidence; contextual interpretation; strengths, limitations, and assumptions; conclusion; APA in-text citations/references; and a section pointing to visible AI-work records. Cross-check every claim against code/output and caption every figure.
Common errors and representative source task	The retained brief requires a comparative real-data report and APA evidence. Avoid screenshots without explanation, unsupported “best model” claims, missing units/sources, confusing correlation or fit with causation, hidden limitations, inaccessible deployment, fabricated references, and undisclosed AI contributions.
Support, extension, and student evidence	Support: report outline, methods sentence frames, evidence-to-claim organizer, screenshot/caption checklist, APA example, and conference feedback. Extension: reconcile conflicting fit criteria, discuss sampling bias/generalizability, or propose improved data collection. Evidence: final report, source list, appendices/figures, shareable model, code/test record, contribution statement, and visible AI documentation.

N3 practice and assessment	Practice begins with a teacher-provided valid model, then parameter stress tests, a small verified upload, comparison rehearsal, and report peer review. Assessment is the digital C8–C10 project and APA-style comparative report; no additional written exam is required by the retained authoritative brief. Feedback uses criterion-linked code/interface testing, model-review comments, and report conference; students submit an error log and remediated version.
Table-free "I can" exit ticket	*I can* keep an interactive PDF valid while shading three event types; clean data and justify estimated candidate models; and connect a screenshot/result to a sourced claim, limitation, and conclusion.
Independent study and required evidence	Maintain a build journal with dated tests, failures/fixes, source checks, AI interactions, and one paragraph explaining each modeling choice. Submit working/shareable output, source code, upload/cleaning record, histograms and fitted curves, comparison evidence, APA report, and correction log.
DUA and extension across N3	Provide starter code, keyboard-accessible controls, high-contrast plots, sample valid inputs, project milestones, parameter/cleaning/report templates, oral screen-record alternative plus required written evidence, and frequent checks. Extend through additional candidates, sensitivity/assumption tests, accessibility improvements, or limitations/generalizability analysis.
Material readiness and risks	The three SharePoint URLs are external and were not verified as accessible; the three relative CSV paths do not resolve to files in Grade 11 Term 1 (repository copies elsewhere do not make the legacy relative links valid here). Hosting/GitHub Pages, internet access, accounts, public-sharing permission, browser compatibility, AI-conversation link visibility/privacy, and APA-source access remain deployment decisions. A teacher must provide or authorize verified datasets and repair links before C9/C10 begin.

6. Learning Evidences, Feedback, and Progression

First Learning Evidence – Current cumulative C1–C4 written evidence	
Prior practice	All four current activities, with compact C1 handout only for reinforcement and selected cumulative prompts rehearsed without embedded solutions.
Assessed criteria and product	C1–C4; a cumulative written response containing classifications/justifications, normalized PDFs and geometric areas, circular-arc/segment reasoning, exponential calculations, and suitability judgments.
Source accuracy and structure	G11_T1_C1C2C3C4_exam0_solution.tex is one assessment-style solved set with embedded solutions, not a separate unsolved current exam. It cumulatively moves from family recognition to uniform/triangular normalization, circular-arc density and segment area, then rate/wait calculations and model evaluation. A teacher-created solution-hidden version is required for administration.
Feedback, error review, and remediation	Return criterion-coded comments; students classify each error as concept, representation, normalization/area, units/formula choice, calculation, interpretation, or assumption. Conduct short method conferences, use the solved logic only after first attempt, then require one parallel correction per affected criterion and a written "why the revision works." No grading threshold is inferred from the repository.
Submitted evidence and next connection	Original response, teacher feedback, error audit, corrected solutions, and C1–C4 reflection. Successful area/interpretation habits prepare the optional normal extension; model-suitability language prepares data fitting.

Second Learning Evidence – Planned C5–C7 extension assessment	
Prior practice	Adapted unsolved z-table, contextual standardization, and inverse-normal tasks drawn from the retained activities, with normal-table.tex as reference.
Assessed criteria and product	C5–C7; a planned written extension assessment with shaded standard-normal regions, table logic, contextual z-transformations, percentiles/cutoffs, central intervals, units, and interpretations.
Readiness	Only the legacy old_G11_T3_C1C7_midterm.tex and paired solution exist; the question paper is mislabeled and covers earlier C1–C4 content as well. It is a reference, not a current assessment, and must be adapted. No current unsolved C5–C7 assessment was found.
Feedback, error review, remediation, evidence, and next connection	Use region-first feedback, table-coordinate checks, standardization/inverse templates, and forward verification of cutoffs. Submit original, annotated curve/table work, error categories, corrected parallel items, and reflection. These inverse/forward normal controls feed the normal mode in the optional digital project.

Third Learning Evidence – C8–C10 digital project extension	
Prior practice	Validated interactive-model prototype, parameter stress test, verified mini-dataset upload/cleaning rehearsal, histogram/curve comparison, and report-outline peer review.
Assessed criteria and product	C8–C10; interactive uniform, triangular, linear, piecewise, and normal PDFs with below/above/between shading; data upload and cleaning; estimates; density-scaled histograms and multiple fitted curves; justified comparison and probabilities; tested deployed/shareable output; APA-style comparative report; and visible, privacy-appropriate AI-work evidence.
Feedback and error review	Interface test conference, model-fit review, and evidence-to-claim report review. Students log invalid parameters, upload failures, scaling errors, discrepancies, broken links, unsupported claims, and citation issues; remediation requires retest evidence and a revised deployment/report.

Submitted evidence and course connection	Source code, link/output, screenshots, test/cleaning/fit records, estimates and probabilities, comparison, report, APA references, AI record, feedback, and revision log. The product integrates mathematical model construction, empirical evaluation, and defensible economic communication. Dataset/link/deployment dependencies must be resolved before assignment.

References and Material Status

Current Term 1 sources

- **Course map:** README.md.
- **C1 current practice with solutions:** 1-practice_activities/G11_T1_C1_practice_activity1.tex.
- **C2 current practice with solutions:** 1-practice_activities/G11_T1_C2_practice_activity2.tex.
- **C3 current practice with solutions:** 1-practice_activities/G11_T1_C3_practice_activity3.tex.
- **C4 current practice with solutions:** 1-practice_activities/G11_T1_C4_practice_activity4.tex.
- **Current cumulative assessment-style solved set:** 2-learning_evidences/G11_T1_C1C2C3C4_exam0_solution.tex.

Current supplementary and reference sources

- **C1 compact solved reinforcement:** 4-extra_material/G11_T1_C1_practice_activity1_print1_solved.tex.
- **Standalone standard-normal reference for the retained extension:** 4-extra_material/normal-table.tex.

Legacy sources actually used as extension references

- **C5 retained practice:** 1-practice_activities/old_G11_T3_C5_practice_activity.tex.
- **C6 retained practice:** 1-practice_activities/old_G11_T3_C6_practice_activity.tex.
- **C7 retained practice:** 1-practice_activities/old_G11_T3_C7_practice_activity.tex.
- **Legacy assessment reference requiring adaptation:** 2-learning_evidences/old_G11_T3_C1C7_midterm.tex.
- **Legacy solution reference requiring adaptation:** 2-learning_evidences/old_G11_T3_C1C7_midterm_solution.tex.
- **Legacy C8–C10 project brief requiring adaptation and dependency resolution:** `2-learning_evidences/old_G11_T3_L4_C8C9C10_activity.tex`.